PortIN — Intelligent Maritime Freight Intelligence Platform

Official SIH 2026 Problem Statement 26006 | SAIL & Ministry of Steel | Generated: 2026-09-03 11:59:20

1. Executive Summary

Executive chartering analysis for 70k MT coal parcel into Paradip.

2. Operational Chartering Parameters

Parameter	Evaluated Value	Source Provenance
Cargo Commodity	Coking Coal	SAIL Blast Furnace Specification
Parcel Size	70,000 MT	Procurement Requirement
Trade Lane	Australia -> Paradip	Major Maritime Bulk Corridor
Recommended Vessel	Handysize	Physical Berth Optimization
Contract Strategy	Short-Term Multi-Voyage (3 Voyages COA)	Volatilty Mitigation Model
Market Signal	MONITOR	HistGradientBoosting Model
Optimal Entry Window	Next 14–21 Days	Forward Freight Curve Analysis
Estimated Total Cost	$3,064,500.00 USD	Aggregated Chartering Model
Operational Risk	24.5 / 100 (LOW)	Multi-Factor Risk Engine

3. PortIN Decision Twin Strategy Evaluation

The Decision Twin ran 36 stochastic perturbations simulating variations across weather, berth queues, and bunker rates. Plan A is recommended as the optimum balance of volume savings and draft safety.

Plan	Strategy	Vessel	Rate ($/MT)	Est. Idle (Days)	Risk
Plan A (Best Overall)	Short-Term Multi-Voyage (3x)	Panamax	$14.15	5.4 days	24.5 (Low)
Plan B (Lowest Risk)	Spot with Guaranteed Berth	Supramax	$15.95	3.3 days	14.0 (Very Low)
Plan C (Lowest Cost)	Long-Term Strategic COA	Capesize / Panamax	$13.05	8.4 days	52.0 (Medium)

4. Data Provenance & Trust Layer

NOTICE: Port constraints and berth physical parameters are based on officially gazetted Port Trust publications. Freight curves are generated via statistical Gradient Boosting models trained on reproducible seeded benchmarks (SIMULATED SIH DEMO DATA). Marine weather data integrates Open-Meteo services for situational awareness. This report is for strategic logistics planning and not for operational vessel navigation or contractual commitment.